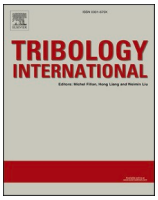
Tribology International 178 (2023) 108080



Contents lists available at ScienceDirect

Tribology International

journal homepage: www.elsevier.com/locate/triboint





Rheology and tribology of nanocellulose-based biodegradable greases: Wear and friction protection mechanisms of cellulose microfibrils

Sergey O. Ilyin *, Svetlana N. Gorbacheva, Anastasiya Y. Yadykova

A.V. Topchiev Institute of Petrochemical Synthesis, Russian Academy of Sciences, 29 Leninsky Prospect, 119991 Moscow, Russia

ARTICLE INFO

Keywords:
Greases
Nanocellulose
Rheology
Tribology

ABSTRACT

Dispersions of microfibrillated cellulose (MFC) in triethyl citrate have been considered as biodegradable lubricants containing 1–7 % MFC. Due to the presence of microfibril entanglements, these dispersions exhibit shear-thinning and shear-thickening behaviors, as well as viscoelastic properties of gels, which makes them typical greases. Moreover, the dispersions maintain their ability to flow at temperatures as low as −50 °C, being potentially applicable as low-temperature lubricants. Microfibrils reduce friction and wear of steel surfaces in two ways. Microfibrils in low concentrations fill irregularities of friction surfaces and decrease their wear, while microfibrils in high concentrations create a smooth chromium-enriched tribological film on the wear surfaces and largely reduce their friction.

1. Introduction

High toxicity and low biodegradability of traditional greases based on mineral oils lead to the fact that such compositions are not safe for the environment. Currently, the transition to biodegradable environmental-friendly materials is one of the priority tasks, and this leads to greater popularity of materials based on renewable, readily available, and biodegradable components [1]. Among such materials, cellulose is especially worth mentioning, as it is the most widespread polymer in the world [2]. The point is that traditional thickeners are metallic soaps, and the extraction of metals (in particular, Li and Al) from mineral raw materials requires large amounts of water and electricity and causes severe environmental pollution. At the same time, cellulose due to its nature has such properties as biodegradability [3,4], renewability [5], non-toxicity [6], and availability [7] along with relatively low cost [8]. As a result, the use of cellulose and its derivatives as additives or thickeners for greases has been gaining popularity in recent decades [9].

The problem is that cellulose particles introduced into traditionally-used base oils form aggregates and tend to sediment [10]. By way of example, nanocellulose currently has found application in the production of greases but only as an additional thickener (an additive) that is introduced in small quantities to existing lubricants rather than as the base for new grease creation [11–14]. Due to instability of native cellulose particles, various cellulose derivatives are used as thickeners to obtain novel biodegradable greases [15]: alkali-treated cellulose [16], epoxidized cellulose [17], methyl cellulose [18,19], ethyl cellulose [20,21], cellulose acetate butyrate [22], stearoyl-grafted cellulose nanocrystals [9], bacterial-treated cellulose pulp [23], and isocyanate-functionalized cellulose [24–26]. Particles of modified cellulose do not agglomerate, allowing obtaining greases on their basis with cellulose acting as the main thickener rather than as an additive. Now, this is the main way to make cellulose-based greases, but it requires chemical modification of the cellulose with additional costs.

However, there is an alternative way to obtain greases based on native cellulose, which consists in selection of base oil of a certain polarity. Indeed, the proper choice of base oil and thickener is important to provide stability of thickener particles against aggregation and sedimentation [27–29]. Furthermore, the type of dispersed particles and their affinity with a continuous medium determine the structure of the resultant mixtures and their rheological and adhesive properties [30–32]. Since cellulose is a polar material, then it should preferably be used with a polar base oil [33,34]. In addition, the oil must be non-toxic, biodegradable, non-volatile, and low-melting in order to obtain fully-biodegradable eco-friendly grease that is effective over a wide range of temperatures. Among all oils, triethyl citrate (TEC) has optimal properties, including low-temperature crystallization at −55 °C [35]. The latter is especially important, since biodegradable low-temperature greases based on cellulose fibers have not yet been developed.

Depending on its nature, cellulose is divided into microcrystalline, nanocrystalline, nano- or microfibrillated, and bacterial one [36–39].

* Corresponding author.
E-mail address: s.o.ilyin@gmail.com (S.O. Ilyin).

https://doi.org/10.1016/j.triboint.2022.108080
Received 19 August 2022; Received in revised form 7 November 2022; Accepted 14 November 2022
Available online 15 November 2022
0301-679X/© 2022 Elsevier Ltd. All rights reserved.

The use of cellulose in the nanoparticle form seems to be the preferred way to obtain better cellulose-based greases, since nanoparticles are not prone to sedimentation in a suitable base oil and thicken it more effectively at the same volume fraction [10]. In view of the necessity to minimize the cost of raw materials for preparing greases, it is preferable to choose microfibrillated cellulose (MFC) among other nanocellulose types, owing to the simplicity of its production that consists in the mechanical processing of cellulose pulp [40–43]. It can be expected that with the correct choice of base oil, cellulose microfibrils without any chemical modification would be able to form the spatial network that is typical for greases and thus act as a thickener for creating greases rather than an additive for existing lubricants. However, the applicability of greases is determined by their rheology [44,45]. There are many works on the rheology of aqueous dispersions of nanocellulose of various types, including microfibrillated cellulose, which shows viscoelastic, shear-thinning, and thixotropic behavior in dispersions that have a yield stress [46–53]. However, there are almost no studies on the rheology of nanocellulose in organic media, and it is not clear whether these dispersions can reduce friction and wear of rubbing bodies. In particular, it has previously been shown that the addition of up to 0.9 % of nanocrystalline cellulose to engine oil reduces friction and wear but does not result in the formation of a grease [54]. The addition of up to 5.2 % of nanofibrillated or nanocrystalline cellulose to castor or sunflower oil also does not produce a grease (a lubricant that has a yield stress) and has an ambiguous effect on wear, resulting in its increase or decrease without a clear correlation with cellulose concentration [55]. Thus, now there are no greases based on unmodified nanocellulose, including low-temperature ones and containing microfibrillated cellulose, as well as the mechanism of action of nanocellulose particles on the tribofilm formation, friction, and wear remains unclear.

On this basis, the purpose of this study is to obtain biodegradable greases by using native microfibrillated cellulose as a thickener and triethyl citrate as a base oil with the examination of their rheological and tribological properties along with the evaluation of the effect of nanocellulose particles on the friction surface.

2. Experimental

2.1. Materials

Microfibrillated cellulose (MFC) Celish KY100S (Daicel FineChem, Japan) was used as a thickener for greases. This nanocellulose was supplied as a 25 % aqueous dispersion that was added to triethyl citrate (TEC, Merck, Germany) with the following mixing on a disperser Ultra-Turrax T18 (IKA, Germany) at 24 000 rpm for 5 min at 25 °C. Then, the mixture was kept in a heat oven at 120 °C for 2 h in order to evaporate the water. By the subsequent addition of TEC to the resulting dewatered system, MFC/TEC dispersions containing 1, 3, 5, or 7 wt% of MFC were obtained (Fig. 1). In this case, three samples of each MFC concentration (C_{MFC}) were prepared for further parallel testing with averaging of the resulting tribological and rheological curves. There was no point in using more than 7 % of MFC in the dispersion, as it became non-flowing and resembled wet sand. Photographs of the obtained systems are shown in Fig. 2.

2.2. Methods

A scanning electron microscope (SEM) Phenom XL G2 (Thermo Scientific, the Netherlands) was used to study wear surfaces and obtain images of MFC particles at an accelerating voltage of 15 kV and a pressure of 60 Pa. The 1 % MFC dispersion in TEC was applied to an aluminum substrate, dried in a heat oven at 80 °C for several hours, and then coated with a thin layer of silver on a sputter 108 auto (Cressington Scientific Instruments, UK). The wear surfaces were examined without silver sputtering. A comparative energy dispersive X-ray spectroscopy (EDX) of steel ball surfaces was performed to evaluate changes in their elemental composition due to wear. For this purpose, EDX of wear surface areas of about 0.1 × 0.1 mm^2 was carried out and then the resulting data collected for at least 20 min were compared with those for the wear-free surface of the same specimen.

Rheological investigations were performed using a rotational rheometer Discovery HR-2 (TA Instruments, USA) with a sandblasted cone-plate unit (the cone diameter of 25 mm, the angle between the conical surface and the plate of 2 °). Flow curves were measured in the steady-state regime with a stepwise increase in a shear rate ($\dot{\gamma}$) from 10^{-3} s^{-1} to 3000 s^{-1} at 25 °C. The tests had been stopping when there was wall slip or internal fracture occurrence, which was determined by a sharp decrease in the measured shear stress (σ) at the next stepwise increase in the shear rate. Large amplitude oscillatory shear was applied to investigate the destruction of the grease structure by stepwise increasing the strain amplitude (γ) from 0.01 % to 1000 % at a constant angular frequency (ω) of 6.28 rad·s^{-1} and a temperature (T) of 25 °C. The values of nonlinear storage (G') and loss (G'') moduli were calculated from the value of the first harmonic of the shear stress [56,57]. Frequency dependences of linear storage modulus, loss modulus, and complex viscosity (η^*) were obtained at a small strain amplitude of 0.1 % and a stepwise change in an angular frequency from 0.0628 rad·s^{-1} to 628 rad·s^{-1} at 25 °C. Temperature dependences of effective viscosity (η) were determined in the heating mode at a rate of temperature increasing of 2 °C/min and a constant shear rate of 0.01, 0.1, 1, or 10 s^{-1}. Rheological characteristics were calculated using standard equations [58,59], and the relative error in their measurement was no more than 5 %.

Tribological studies were carried out on a tribometer CETR-UMT (Bruker, USA) by using a ball-on-disc friction pair (ASTM G99-17)

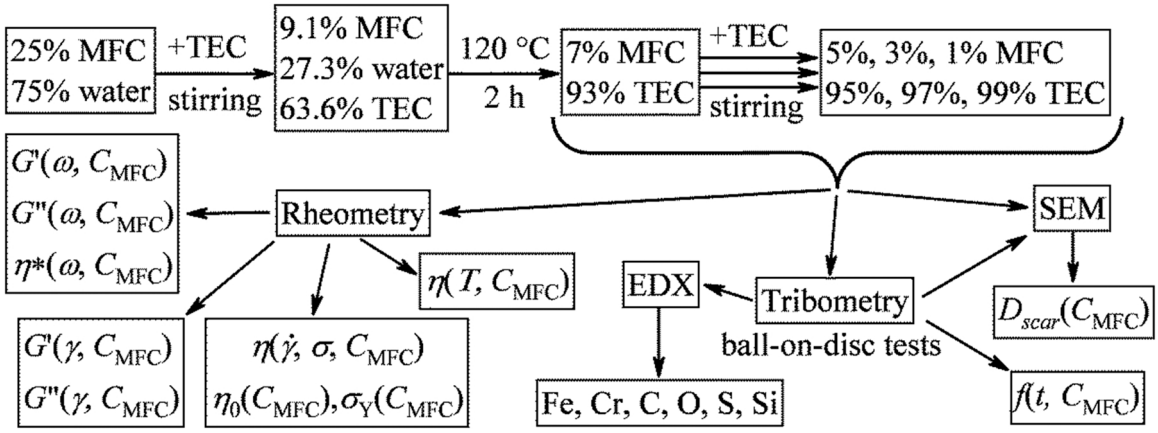


Fig. 1. General scheme of the research workflow.

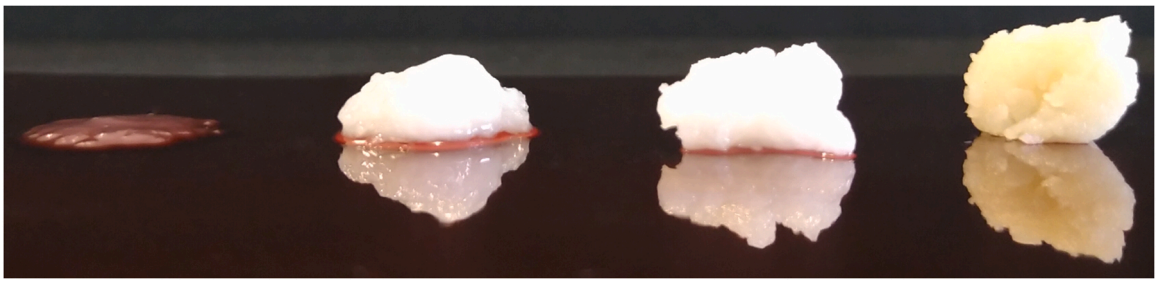


Fig. 2. Photographs of dispersions containing 1, 3, 5, or 7 % of microfibrillated cellulose in triethyl citrate. The cellulose concentration increases from left to right, and the dispersions are placed on the polyimide film.

where the ball draws a circle with a diameter lying within the range of 35–55 mm on the disc that was lubricated with a 1-mm layer of the grease to be tested. A thick grease layer allowed avoiding starvation during the tests [60] because otherwise a seizure would have occurred within a couple of minutes from test starting. Within the process, a ball (440 C steel, diameter of 6.35 mm) moved with a linear velocity of 0.15 m/s under a force of 30 N over the surface of a lubricated steel disc for 20 min. The specified force corresponded to a contact pressure of 1910 MPa, which is close to the maximum pressure in rolling bearings and gears [61]. Under these conditions, boundary friction is realized and greases that work well under high loads are required to prevent wear and seizure. The chosen ball speed is high enough for the moving surfaces to rub down in a reasonable time of about several minutes and for their sufficient wear within 20 test minutes to quantify the wear confidently (each ball passed about 1000–1600 rounds on its circular trajectory on the moving disk). The evaluation of wear was done based on ball wear because the balls were subjected to the same exposure in all experiments, unlike the discs (in the parallel experiments, the trajectory diameter of the balls was varied rather than their linear velocity). The surfaces of balls and disks differed in surface roughness (60 nm and 280 nm respectively), but this did not actually matter since the friction in the boundary conditions under high loads and over long periods of time leads to high wear and negligible effect of the original roughness. The temperature of tribological tests was taken similar to that of the rheological ones and was equal to the standard ambient temperature of 25 °C. The friction coefficient (f) was calculated according to the following equation:

$$f = (F_x^2 + F_y^2)^{0.5}/F_z, \quad (1)$$

where F_x, F_y, and F_z are the average lateral, tangential, and normal forces respectively. The diameter of the wear spot on the ball surface (D_{scar}) was determined by using SEM-images. For each MFC concentration, the tests were performed three times and the average characteristics were calculated.

3. Results

3.1. Morphology and rheology of MFC dispersions

Cellulose microfibrils form entangled agglomerates, according to SEM data (Fig. 3a). In this respect, cellulose microfibrils resemble crystals of lithium soap, which also form entangled fibers in a given medium (Fig. 3b). However, cellulose microfibrils are much smaller in diameter and consequently are able to bend and entangle with each other more frequently than the fibers from lithium soap. Apparent diameter of individual cellulose fibers ranges from 0.1 μm to 1 μm, whereas their length can reach up to several tens or even hundreds of micrometers. Associationally, it can be noted that the spatial structure of microfibril dispersion resembles polymer melt structure, where microfibrils play the role of macromolecular chains that are entangled with each other like macromolecules. On this basis, it should be expected a certain similarity of the rheological behavior of microfibrillated cellulose dispersions with that of polymer melts.

Indeed, all MFC dispersions in TEC are non-Newtonian fluids with complex rheological behavior (Fig. 4a). The dynamics of these dispersions can be interpreted in terms of the microfibrils themselves or their agglomerates—flocs [50]. In the field of very low shear rates ($< 5{\cdot}10^{-5}$–$5{\cdot}10^{-4}$ s^{-1}), the effective viscosity of the dispersions is practically independent on shear rate, which can be associated with their deformation without destroying the structure. In analogy with the rheology of solutions and melts of polymers, such behavior occurs when the destruction rate of contacts between cellulose microfibrils (entanglements) is lower than their recovery rate and their total number in the deforming process remains constant. Alternatively, it can be contacts between the flocs. In the field of higher shear rates ($> 5{\cdot}10^{-4}$ s^{-1}), the effective viscosity monotonically decreases with the increase of shear rate, which indicates the ordering of cellulose microfibrils (their orientation in the shear field) or a decrease in the size of the flocs [62]. It can be supposed that in semi-dilute dispersions, which contain a floc-free volume where flocs can float and disagglomerate with a decrease in the effective hydrodynamic diameter, their behavior would be dictated

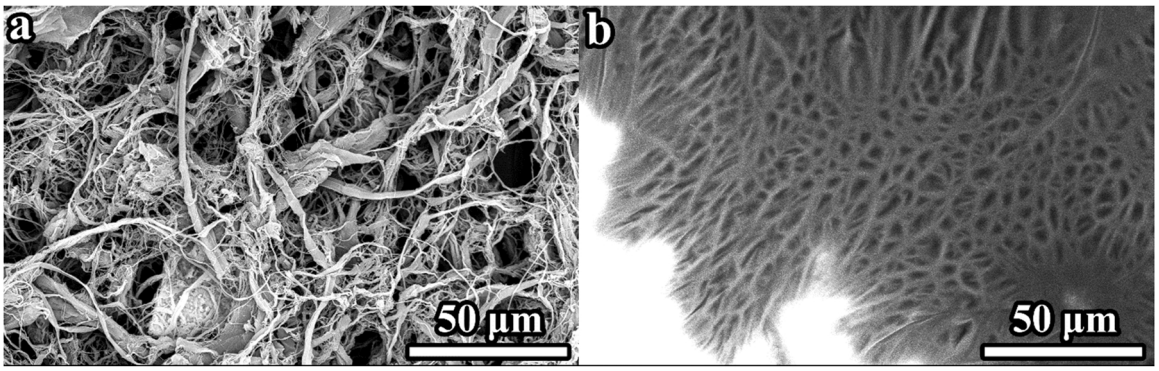


Fig. 3. SEM-images of microfibrillated cellulose (a) and lithium 12-hydroxystearate (b) after drying their 1 % dispersions in triethyl citrate.

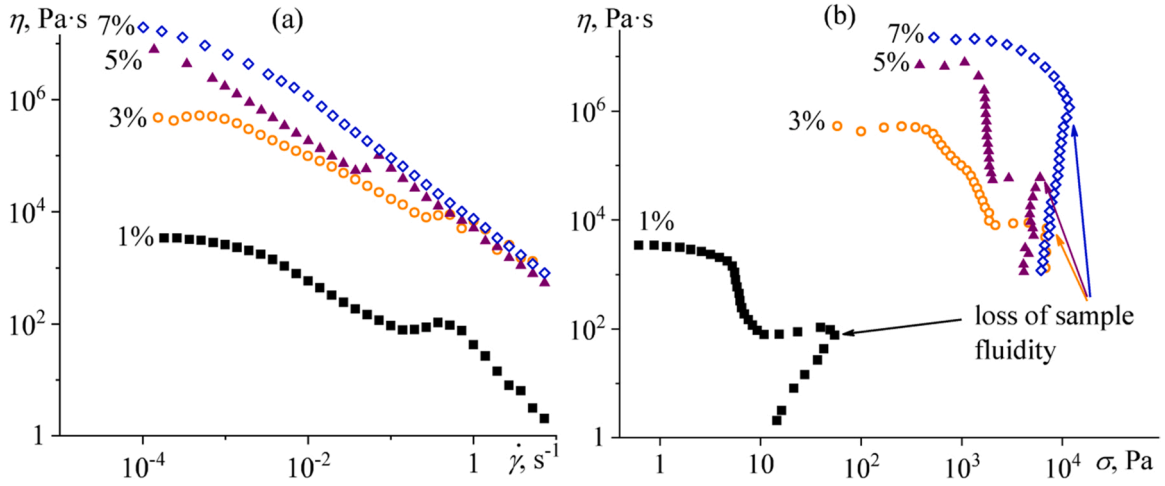


Fig. 4. Dependences of the effective viscosity on the shear rate (a) and shear stress (b) for dispersions containing 1, 3, 5, or 7 wt% of microfibrillated cellulose in triethyl citrate at 25 °C.

by the dynamics of these flocs. On the contrary, microfibrils and their entanglements would determine the rheology of concentrated systems, since they do not contain the free volume necessary for transformations of flocs. In our case, we can only speculate that the nominal boundary between semi-dilute and concentrated systems is between 1 % and 3 % MFC, because these systems have very different viscosities, while the viscosities of 3–7 % dispersions differ from each other notably less.

At moderate shear rates (> 0.05–$0.2\ s^{-1}$), the effective viscosity of dispersions slightly increases, but only in the case of the systems containing 1–5 % of MFC. This behavior may be related to the reorganization of dispersion structure due to the approach of oriented flocs or microfibrils to each other and the intensification of interactions between them [63]. Subsequent decrease in the effective viscosity at higher shear rates (> 0.1–$0.5\ s^{-1}$) can be explained by the continuation of decrease in floc size or further microfibril orientation. Nevertheless, such experimental data interpretation does not explain the absence of shear thickening behavior for dispersion containing 7 % of MFC.

An additional explanation of what happens during the deformation of dispersions can be also obtained by redrawing the flow curves as a function of the effective viscosity versus shear stress (Fig. 4b). All samples have a yield stress: a narrow stress range where the effective viscosity drops by a factor of tens [64]. The existence of the yield stress is characteristic for greases [45] and connected with the destroying of their continuous network structure [65,66]. In our case, microfibril entanglements form this structure both in semi-dilute and concentrated systems. The difference is that all entanglements are evenly distributed in the volume of the concentrated dispersions (3–7 % MFC) and play the equal role, whereas microfibril entanglements between individual interconnected flocs are only important for the semi-dilute system (1 % MFC). In the concentrated dispersions, as the shear rate increases, microfibrils are orienting in the flow direction, and this causes a decrease in the number of entanglements and consequently a drop in the effective viscosity. In the semi-dilute system, flocs detach from each other and move freely in the dispersion medium with a gradual decrease in the size as the shear rate increases. At the same time, the dilatant growth of effective viscosity may be associated with the tightening of microfibrils within the knots of entanglements [67], which can also bind individual flocs together into a coherent whole. At low shear rates, microfibrils slip free through the entanglement knots, whereas high shear rates lead to the friction growth in the entanglement knots and the cessation of microfibril movement through these knots. This behavior results in the fact that the number of entanglements stops decreasing with increasing shear rate and even starts to grow, because the entanglements do not disentangle due to the microfibrils stuck in them. This eventually results in the growth of dispersion effective viscosity with the elevation of shear rate.

At the same time, the friction growth in entanglement knots as the shear rate increases leads to more and more knots becoming tight. As a result, the system loses an ability to flow at a certain shear rate and breaks down with the loss of continuity. The latter is confirmed by the visual observation of deforming system (the part of sample leaves the measuring gap between the cone and the plate) and by the reduction of a shear stress as the shear rate increases (instead of growth; shown by arrows in Fig. 4b). A part of the sample breaks and is squeezed out of the measuring gap, while the remaining part likely wall slips along the measuring surfaces of the rheometer [68]. For polymer melts, the loss of continuity does not occur at high shear rates because of higher elasticity. Although high deformation rates in fact lead to a loss of fluidity of polymers, this loosing is temporary due to their elasticity, manifesting itself in the arising of instability caused by the periodic transition between the flow of polymer melt and its wall slip—so-called stick-slip behavior [69]. At the increase of MFC concentration in TEC, the yield stress of dispersion increases, which leads to the absence of shear thickening behavior of 7 % dispersion due to the high concentration of tightening entanglements resulting in the easy loss of fluidity of this sample. In turn, the loss of fluidity leads to wall slip and a substantial error in estimating the effective viscosity of the dispersion [70].

Thus, the loss of fluidity actually limits the maximum shear rate in viscosity measurements to about 0.1–$10\ s^{-1}$, depending on grease consistency, and the value measured at higher shear rates is not viscosity but some effective value that has no physical meaning. At the same time, the shear rate between bodies rubbing in the boundary or mixed mode can reach magnitudes as high as $10^6\ s^{-1}$. Even if there were no wall slip, this high shear rate cannot be realized on a rotational rheometer allowing at best a shear rate of about $10^4\ s^{-1}$. In this respect, rheological tests of greases operating under boundary conditions are needed not to predict their friction behavior (as for greases operating under hydrodynamic lubrication), but rather to draw conclusions about their structure. Further information about the features of greases can be obtained by measuring their viscoelasticity via small-amplitude oscillatory shear that does not destroy their structure.

All of the examined MFC dispersions show viscoelasticity (Fig. 5). At low strains (in the linear area of mechanical behavior), the storage and loss moduli do not depend on strain amplitude, while the storage modulus significantly exceeds the loss one (Fig. 5a). In this regard, MFC dispersions behave more like solids [71]. As the strain amplitude increases above a certain critical value (at the transition in nonlinear area), the storage modulus starts to decrease, while the loss modulus

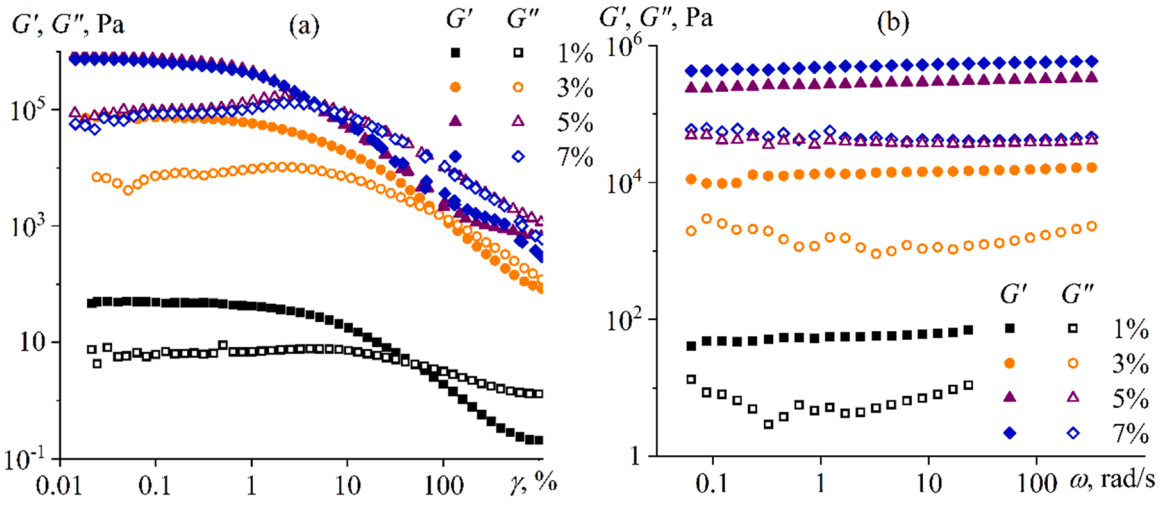


Fig. 5. Amplitude (a) and frequency (b) dependences of storage modulus (filled symbols) and loss modulus (open symbols) for dispersions containing 1, 3, 5, or 7 wt % of microfibrillated cellulose in triethyl citrate at 25 °C.

increases with passing through a local maximum and then also decreases. As a result, the loss modulus becomes more than the storage one, i.e. dispersions start to behave more like liquids.

The transition from solid-like behavior to liquid-like one occurs at a strain of about 1 % and is associated with the destruction of the entanglement structure of microfibrils. In the case of macromolecular entanglements, the transition from linear behavior to nonlinear one takes place at larger strains reaching approximately 25 % and is not accompanied by passing the loss modulus through a local maximum [72]. Probably, the smaller extent of the linear viscoelasticity region for MFC dispersions is due to the lower flexibility of microfibrils in comparison with macromolecular chains. Meanwhile, the presence of a local maximum of loss modulus (so-called weak strain overshoot [73]) can be associated both with the increased interactions between microfibrils due to their orientation and with an increased number of entanglement knots due to their tightening.

In the linear area of mechanical behavior, the storage and loss moduli practically do not depend on deformation frequency (Fig. 5b). Moreover, the storage modulus exceeds the loss modulus over the entire frequency range and the difference between the moduli approximately is equal for all dispersions. This behavior points out gel-like behavior of these samples [74], and the knots of the gel network correspond to the knots of the microfibrillar entanglement network. In a similar way, macromolecular entanglement network gives polymers high-elastic (rubbery-like) properties in a certain field of temperatures and observation times [75]. As the MFC concentration increases, the entanglement number grows and consequently the storage and loss moduli of dispersion rise.

There is the Cox-Mertz empirical rule for polymer melts and solutions, according to which the steady-state viscosity coincides with the complex viscosity when the shear rate and the angular frequency are numerically the same [76]. This rule works well for homogeneous glass-forming or polymer systems but is broken if the viscoelasticity is partially or fully caused by the dispersed phase [77–79] . The rule breaks in our case, and the complex viscosity significantly exceeds the steady-state viscosity (Fig. 6a). This is because the complex viscosity is measured at small strains, when dispersion structure is in an unbroken state and has a maximum number of entanglements that disappear and arise at the same rate. In contrast, the steady-state viscosity is determined under the condition of sample flow when entanglements are

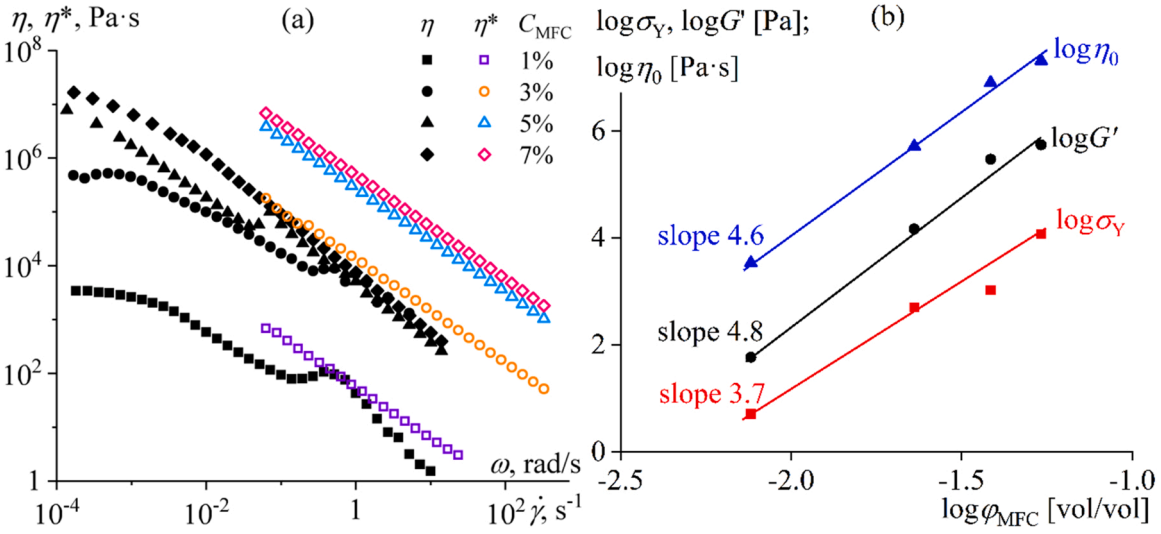


Fig. 6. Dependences of the steady-state viscosity (filled points) and complex viscosity (hollow points) on the shear rate and angular frequency respectively for microfibrillated cellulose dispersions in triethyl citrate at 25 °C (a), as well as dependences of the low-shear viscosity, yield stress, and storage modulus of these dispersions on the volume fraction of microfibrillated cellulose (b).

continually disappearing and forming again, but the disappearance rate increases as the shear rate elevates, while the forming rate remains constant. With increasing the shear rate, this leads to a decrease in the average number of entanglements existing at a moment in time and a corresponding decrease in the steady-state viscosity. It is interesting that because of dilatant growth, the steady-state viscosity of low-concentrated systems (with 1–3 % MFC) reaches or nearly reaches values of their complex viscosity. In other words, the tightening of microfibrils within the entanglement knots with the cessation of their disentanglement leads to the growth (or rather restoration) of the total number of the knots to the level characteristic of the system undestroyed by continuous shear.

Let us consider the concentration dependences of the storage modulus, yield stress, and low-shear viscosity (Fig. 6b). All these indicators are linearized in log-log coordinates and become higher with increasing the MFC volume fraction (φ_{MFC}), whose logarithm is formally negative since φ_{MFC} lies within 0.0076–0.054 (corresponds to a MFC mass percent of 1–7). The slopes of the concentration dependences of the low-shear viscosity and storage modulus are about the same: 4.6–4.8. Interestingly, η is proportional to $\varphi^{4.67}$ for concentrated polymer solutions that are characterized by the presence of macromolecular entanglements and are prepared in θ-solvent [80]. Thus, the rheological behavior of cellulose microfibrillar dispersions resembles to some extent that of polymer solutions in a neutral solvent, in which there is no unfolding or collapsing of macromolecular coils.

The fundamental difference between dispersions of entangled microfibrils and solutions of entangled macromolecules is the yield stress behavior of the former. In this case, the yield stress value increases with increasing the MFC concentration in a less intensive way than the low-shear viscosity and storage modulus: $\sigma_Y \sim \varphi^{3.7}$. It can be noted that for nanocellulose with spherical-like particles, the yield stress and storage modulus depend less intensely on the cellulose concentration in TEC ($\sigma_Y \sim \varphi^{1.9}$, $G' \sim \varphi^{2.5}$), possibly due to the absence of entanglements between the spherical-like particles [33].

Important factors for greases are the weak dependence of the effective viscosity on temperature and the ability to flow in very cold conditions. The viscosity of pure TEC increases by 8 decimal orders of magnitude at cooling from 20 °C to its crystallization temperature, which can be as low as −65 °C due to supercooling of this liquid (Fig. 7). At such high effective viscosity as 10^3–10^6 Pa·s, pure triethyl citrate cannot serve as a lubricant: in practical terms of short observation times, it behaves like a solid body rather than a liquid, since the pour point of a glass-forming liquid approximately corresponds to its reaching a viscosity of about 300–1000 Pa·s [81]. This limits the lower boundary of the operating temperature range for pure TEC at −48 °C. Thus, the low melting temperature of base oil (−55 °C in our case) does not mean that it can function as a lubricant at equally low temperatures, as ester oils need a branched structure to be low viscous at low temperatures [82].

The addition of MFC into TEC results in a weakening of the effect of temperature on the viscosity properties, which, however, appear to be dependent on the shear rate. For example, the effective viscosity of 1 % MFC dispersion during its cooling changes within 2.5–4 decimal orders of magnitude, depending on the applied shear rate. At the same time, the temperature at which this dispersion reaches an effective viscosity of about 1000 Pa·s is approximately −50 °C (under high shear rates that are actually observed in the gap between the rubbing bodies). An increase in the MFC concentration up to 5 % leads to a change in the effective viscosity within 3 decimal orders with decreasing temperature, while the assumed temperature of flow loss (reaching the effective viscosity of 1000 Pa·s) increases up to −40 °C. Therefore, the addition of MFC makes the viscosity properties of greases less dependent on temperature but does not reduce significantly their minimum possible operating temperature.

3.2. Tribology of MFC dispersions

Antifriction properties of greases are determined by their ability to confront the contact of two rubbing surfaces at a specified load. Let us consider the effect of the MFC concentration on the friction coefficient when a steel ball slides on a steel disc lubricated with an MFC/TEC mixture (Fig. 8, the data were averaged based on several tests).

At the lubricating of rubbing surfaces by pure TEC, the friction coefficient decreases during first 4 test minutes due to the lapping of the surfaces and then takes approximately a constant value (equal to 0.095 after 20 min from the beginning of the test, see Table 1). The introduction of 1–3 % of MFC into TEC does not lead to a noticeable improvement in its antifriction properties but reduces the lapping time of the surfaces fourfold (down to 1 min) and also minimizes friction during the lapping. A further increase in the MFC concentration up to 5–7 % decreases the steady-state friction coefficient down to 0.071–0.077, but the lapping process undergoes a transformation. In the case of 7 % MFC concentration, the friction coefficient passes through a well-defined maximum after 1 min from the starting of the test, while the entire lapping process requires more time—about 4 min. As a result, this changing of lapping mechanism allows for a more efficient

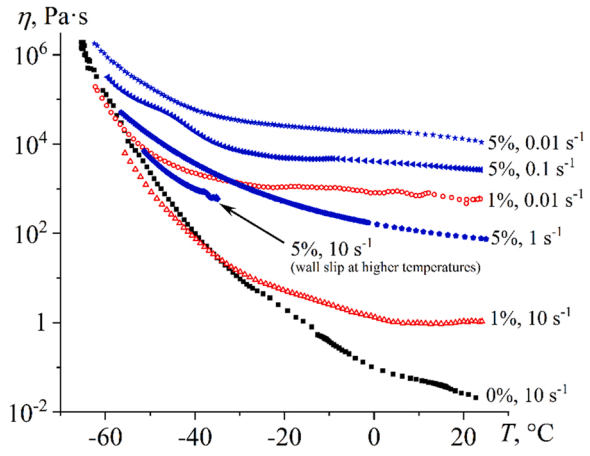


Fig. 7. Temperature dependences of the effective viscosity for triethyl citrate containing microfibrillated cellulose (MFC). The MFC concentration and shear rate are indicated near the curves.

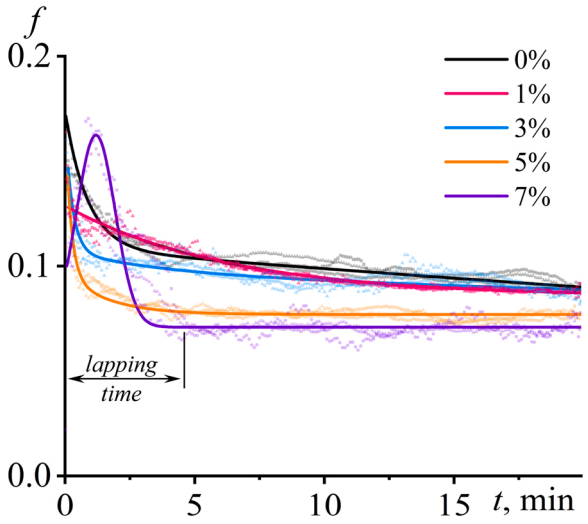


Fig. 8. Time dependences of friction coefficient for pure triethyl citrate and greases containing microfibrillated cellulose with the concentration indicated in the legend. The semi-transparent dots show the experimental data of three parallel experiments for each concentration, while the solid lines represent the result of their averaging.

Table 1
Consistency, friction coefficient, and wear scar diameter when using different triethyl citrate-based dispersions as lubricants.

Lubricant	*p*, mm/10	NLGI number	Oil separation, %	*f*[1]	D_{scar},[1], μm
TEC	–	–	–	0.095 ± 0.019	645 ± 7
1 %MFC+TEC	465 ± 15	000	>70	0.090 ± 0.009[b]	444 ± 13[a]
3 %MFC+TEC	230 ± 10	3	17 ± 1.5	0.092 ± 0.017[b]	515 ± 4[a]
5 %MFC+TEC	170 ± 10	5	7.0 ± 0.4	0.077 ± 0.014[a]	491 ± 9[a]
7 %MFC+TEC	90 ± 5	6	2.3 ± 0.2	0.071 ± 0.025[a]	739 ± 12[a]
13 % $C_{18}H_{35}LiO_3$ +TEC	280 ± 10	3	21 ± 2	0.090 ± 0.016[b]	575 ± 14[a]

[1] The letter after the experimental data shows the p-value—the probability that the addition of MFC did not actually affect this parameter [83]: [a] $p < 0.05$ (there is evidence that it does not), [b] $p > 0.05$ (there is no evidence).

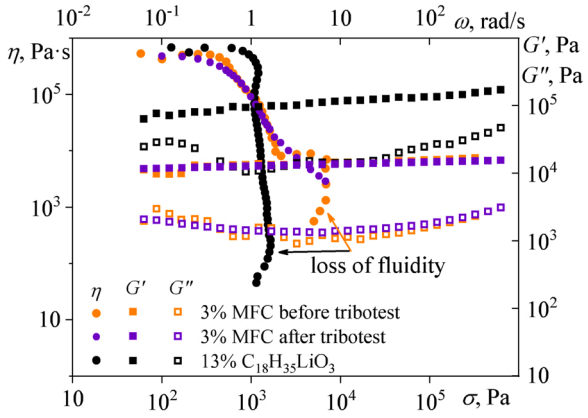


Fig. 9. Dependences of effective viscosity on shear stress (filled circles) and storage and loss moduli on angular frequency (filled and open squares respectively) for dispersions containing 3 wt% of microfibrillated cellulose or 13 wt% of lithium 12-hydroxystearate in triethyl citrate at 25 °C. Data for cellulose-based grease are presented before and after it passes the 60-min ball-on-disc tribotest.

reduction of the friction coefficient that also is less subject to fluctuation over time.

In our case, there is no visible correlation between the rheological characteristics and the tribological performance of the greases. This is quite expected since rheological parameters play a key role in tribology when lubricants are used in hydrodynamic mode when their viscosity is essential [84]. In the boundary lubrication mode, the influence of the rheological features of greases is not apparent, but they determine the consistency of greases and consequently their ability to be used in particular applications, regardless of the specific friction mode. An increase in the MFC concentration makes the NLGI grade of the greases higher (Table 1). In fact, the use of 1 % MFC is not sufficient to create a grease because the resulting product has a liquid consistency and does not even pass the dropping point and oil separation tests. The situation changes when 3 % MFC is added, resulting in a grease of normal consistency with a NLGI grade of 3 and an oil separation degree of 17 % under pressure of 100 kPa for 30 min. A higher MFC concentration of 5 % or 7 % thickens the grease severely so that its NLGI grade becomes 5 or 6 respectively, limiting its use to specific tasks. The colloidal stability of these greases is also strongly increased to 7 % and 2.3 % respectively. In all cases, the dropping point of greases containing 3–7 % MFC exceeds at least 155 °C that is the flash point of triethyl citrate.

In order to understand the effectiveness of MFC as a thickener, let us compare it with the common lithium 12-hydroxystearate thickener, which creates a grease with an NLGI grade of 3 when added at 13 % concentration to the TEC. The consistency of this grease matches that of a dispersion containing 3 % MFC (Table 1). The difference in the concentrations of lithium soap and cellulose microfibrils to achieve the same thickening effect is due to the different thickness of soap and cellulose fibers: cellulose microfibrils are thinner and as a result are able to bend better and form a denser structural network with the same volume fraction (see Fig. 3). Consequently, much less cellulose microfibrils are needed to achieve a certain consistency with the same base oil. If to compare their flow curves (Fig. 9, filled circles), both systems have practically the same low-shear effective viscosity and the value of the yield stress (about 10^3 Pa). A feature of the dispersion thickened with MFC is the presence of a step on the flow curve, which is probably due to the shear-thickening behavior of microfibrils tightening in the entanglement knots. In addition, both systems are gels ($G' > G''$, $G' \approx const$), although the one containing lithium soap is stiffer. This means a larger number of contacts between the thickener particles, which is to be expected since this grease contains more of them (13 % vs. 3 %). Thus, MFC as a thickener allows achieving the similar rheological properties (yield stress, gel-like behavior) but at a much lower concentration, most likely due to high anisometricity of microfibrillar particles. Furthermore, the values of the friction coefficient turn out to be comparable for both systems as well (about 0.09, Table 1). Moreover, the cellulose-based grease has also a sufficiently good mechanical stability; at least its rheological properties practically do not change after 60 min of operation between the sliding ball and the disk (see Fig. 9).

However, the antiwear activity of greases is also important and deserves to be studied. The effect of MFC on the lapping and consequently the wear is confirmed by data on the size of wear scar formed on the surface of the moving ball. The addition of as little as 1 % MFC leads to a significantly decrease in the wear scar diameter, which however becomes larger as the MFC concentration increases (Table 1). Moreover, the wear scar diameter at 7 % concentration of MFC exceeds that obtained with using pure TEC as a lubricant. Such behavior of these greases probably is due to the high hardness of cellulose microfibrils, which is comparable to the hardness of steel [85]. The more the MFC concentration, the more is the wear scar diameter as a result. Nevertheless, when the concentration of MFC is within 1–5 %, the wear scar diameter is smaller than with using the system thickened with lithium soap (Table 1).

Thus, MFC does reduce the wear well when used in combination with TEC, which should be based on some mechanism. The introduction of solid particles of a thickener into a base oil most often is due to their ability to reduce friction, but the wear of rubbing surfaces nevertheless arises in one way or another. To determine the type of wear (adhesive or abrasive) that occurs during the operation of cellulose microfibrillar greases, let us consider SEM images of the wear surfaces.

When pure base oil is used as a lubricant, the surface of the wear spot is smooth but contains microscopic potholes (Fig. 10a). In this respect, there is no sign of abrasion wear as there are no grooves on the surface. The introduction of 1 % MFC into the oil leads to a significant reduction in the diameter of the wear spot and the disappearance of potholes on its surface (Fig. 10b). At the same time, it is possible to notice very faint parallel stripes on the surface, which become more pronounced when using 3 % or 5 % MFC (Figs. 10c, 10d). It is possible that these are traces of some mild abrasive wear caused by cellulose particles rather than metal ones. In any case, the wear spot diameter becomes slightly larger. However, a truly great transformation in the appearance of the wear surface is caused by 7 % MFC: the surface becomes much larger in area, with wider parallel stripes, and the presence of massive potholes (Fig. 10e). As a way to understand the situation, high-magnification SEM images can be used to inspect the details of defects on the wear surfaces.

The use of pure TEC as a lubricant leads to the defect appearance on

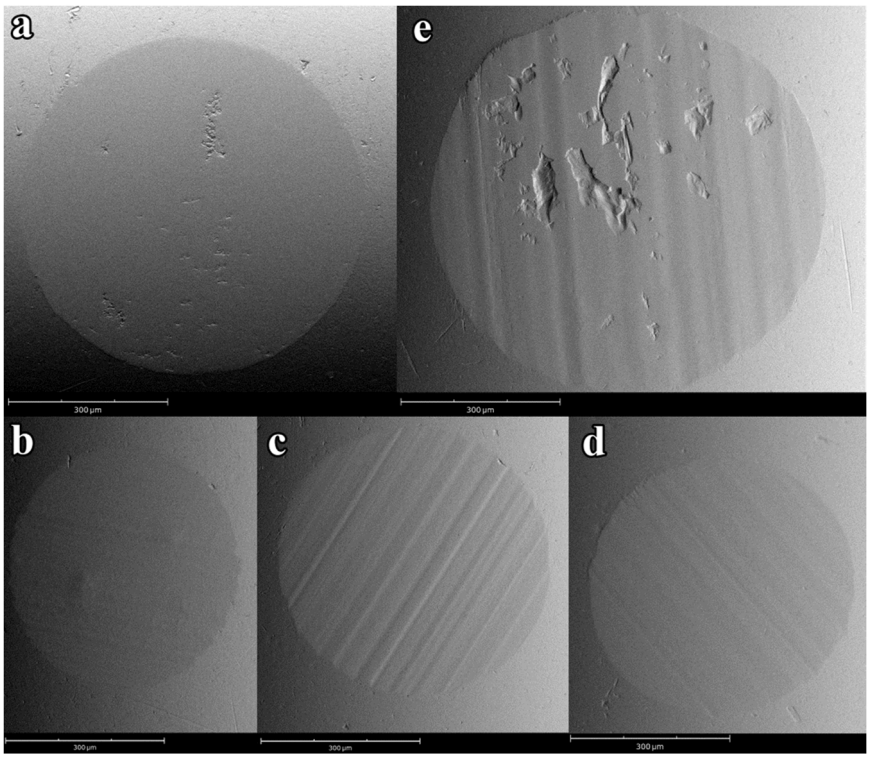


Fig. 10. SEM-images in topographical mode of balls' wear spots after ball-on-disc tests with using pure triethyl citrate (a) or greases containing 1 % (b), 3 % (c), 5 % (d), or 7 % (e) of microfibrillated cellulose.

the rubbing surface such as burrs and furrows (Fig. 11a). Moreover, the surface exfoliation is observed, but there are no grooves that remain after abrasive wear [86–88]. Such surface damages indicate that the consequence of using pure base oil as a lubricant is adhesive wear [89, 90]. Perhaps the absence of abrasive wear is caused by the openness of the ball-on-disc tribological cell, which allows the formed metal particles to leave the wear zone by being squeezed out by the ball.

At the addition of 1 % MFC into the base oil, the pattern changes

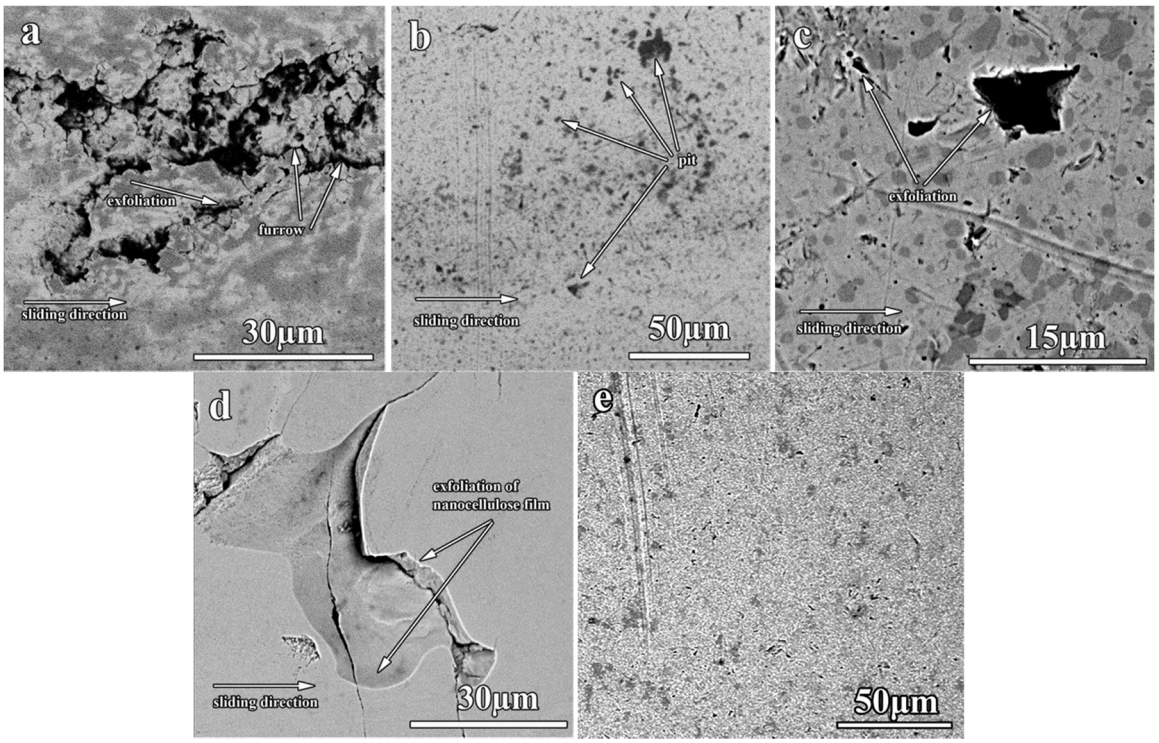


Fig. 11. SEM-images in compositional mode of balls' surfaces after ball-on-disc tests with using pure triethyl citrate (a) or greases containing 1 % (b), 5 % (c), or 7 % (d) of microfibrillated cellulose in comparison with the original wear-free ball surface (e).

(Fig. 11b). The depth of furrows and surface irregularities are greatly reduced, but some dark blotches are observed. These blotches are probably either small potholes or pits caused by adhesion wear of steel surface during friction, since they are not present on the original wear-free surface (Fig. 11e).

Further increase of the MFC concentration up to 5 % makes the wear surface more irregular and bumpy (Fig. 11c). Dark spots appear on the surface and resemble potholes, whose sharp boundaries may indicate a detachment of a tribological film in these areas [54]. Perhaps MFC at this concentration forms a thin tribological film, the thickness of which is insufficient to protect rubbing surfaces. Judging by the SEM image, the film thickness hardly exceeds 0.5 μm, while the diameter of microfibrils is in the range of 0.1–1 μm (Fig. 3). It is therefore difficult to expect that the tribofilm originates from the microfibrils themselves, but rather from the products of their abrasion. Nevertheless, there are no abrasion grooves, despite the presence of some parallel stripes visible on low magnification SEM images (Fig. 10). Possibly, their formation is due to stratification of dispersed microfibrils during the flow of grease in the narrow gap between the rubbing bodies [79,91], resulting in the formation of wear strips in places enriched or depleted in microfibrils.

When the MFC concentration is increased up to 7 %, the formation of tribological film is observed more clearly: few places on the wear surface show detachment of the tribological film due to partial breakdown (Fig. 11d). Judging by these fracture places, the film thickness increases up to 5 μm and thus exceeds the diameter of the microfibrils that can now potentially be incorporated into its composition. At the same time, all other surface defects such as potholes, burrs, and furrows completely disappear. As a result, the wear surface has a smoother appearance than at using other greases.

Indeed, if to return to the time dependences of friction coefficient, its lowest value reaches for the grease containing 7 % of MFC (Fig. 8). Moreover, the friction coefficient decreases as the MFC concentration increases, and this is also consistent with the gradual smoothing of the wear surface in accordance with the SEM images.

Evidence in favor of the formation of a special tribofilm under the influence of nanocellulose-based grease is also the data of elemental analysis of wear surfaces (Table 2). Wear-free surfaces (but prelubricated with tested greases) contain primarily iron (77.4 %), chromium (17.9 %), and carbon (3.9 %), along with traces of oxygen (0.4 %), silicon (0.2 %), and sulfur (0.2 %). The chromium and silicon content is quite in line with the expected values for the applied steel grade [92], while the carbon concentration is markedly higher (3.9 % versus 0.96 %). The latter can be attributed to the filling of irregularities of the original steel surface with ester oil, which also results in the appearance of oxygen on the unworn surface. The steel should also contain manganese (0.32 %), but its spectral bands overlap with those for chromium and iron, so that its content cannot be accurately estimated. Sulfur is an undesirable impurity to steel, so that its presence can be regarded as inevitable and permissible.

Table 2
The initial elemental composition of the wear-free surface and its absolute change caused by friction with using dispersions of microfibrillated cellulose in triethyl citrate as greases.

C_{MFC}, wt%	Fe, wt%	Cr, wt%	C, wt%	O, wt%	S, wt%	Si, wt%
-[1]	77.4	17.9	3.9	0.4	0.2	0.2
0[2]	$+0.9^b$	-0.7^b	-0.7^b	$+0.4^b$	$+0.2^a$	-0.1^a
1	$+0.3^b$	-0.2^b	-0.1^b	$+0.1^b$	0.0^b	-0.1^a
3	-0.2^b	-0.3^b	$+0.7^b$	-0.1^b	0.0^b	-0.1^a
5	-2.9^b	$+0.8^b$	$+1.9^b$	$+0.3^b$	0.0^b	-0.1^a
7	-4.9^a	$+1.4^a$	$+1.3^b$	$+2.3^a$	0.0^b	-0.1^a
SD,[3]%	±1.7	±0.6	±1.5	±0.2	<0.05	<0.05

[1] The first data line refers to the initial wear-free prelubricated surface.
[2] The letter after the experimental data shows the *p*-value—the probability that the addition of MFC did not actually affect the content of this element [83]: [a] $p < 0.05$ (there is evidence that it does not), [b] $p > 0.05$ (there is no evidence).
[3] The standard deviation based on the results of averaging 3–5 independent measurements for each concentration.

Wear of the surface lubricated with pure triethyl citrate leads to a noticeable increase in the concentration of iron, oxygen, and sulfur with the opposite decrease in that of chromium, carbon, and silicon. The decrease in the carbon concentration may be due to frictional smoothing of the surface having thus fewer irregularities into which the base oil can penetrate, while the increase in the oxygen content is probably caused by oxidation of the steel's metals by the action of air oxygen and high local temperatures. An intriguing thing is the change in the ratio between chromium and iron in favor of increasing the concentration of the latter. Due to wear, the content of metals usually decreases with increasing oxygen and carbon concentrations [93], and wear reduction is ascribed to the formation of Fe_3O_4 that adheres to the contact surface [94,95]. It seems to be the same case.

The situation becomes more interesting if we consider the evolution of the ratio of the elements when nanocellulose is added to the lubricant: the higher is the MFC concentration, the more chromium is in the tribological film. Moreover, the chromium content becomes even higher than in the original wear-free surface, while the iron concentration drops dramatically. At the same time, the carbon concentration also increases, while the oxygen content passes through a local minimum and then reaches a maximum in the tribofilm formed under the operation with the grease containing 7 % MFC. The increase in the carbon and oxygen concentrations in the latter case can be associated with the inclusion of cellulose macromolecules (or their residues) in the tribofilm composition. What is not clear is the oppositely directed change in the iron and chromium amounts. Although most of the changes in surface composition are not definitely reliable from a statistical perspective (see note to Table 2), this is not the case of using the 7 % MFC grease resulting in the formation of a specific tribological film with statistically significant changes in composition. Possibly, the increase in iron during friction with cellulose-free lubricant is due to its lower reactivity compared to chromium and thus less solubility in the form of resultant compounds (according to the standard reduction potentials [96]). In turn, the greater reactivity of chromium promotes the formation of a nanocomposite chromium-enriched tribofilm involving cellulose. Furthermore, as chromium is harder than iron, its accumulation can also be the cause of changes in friction and wear in addition to the effects of the nanocellulose itself. Chromium increases the stiffness of the surface layer and thus reduces abrasive wear, while cellulose has less adhesion to metals than the metals themselves and its accumulation within the surface layer is supposed to result in less adhesive wear and decreased friction. Since the SEM images suggest adhesive wear rather than abrasive one (see Fig. 11, e.g., lack of grooves), the role of cellulose in improving tribological properties is central.

4. Discussion

Cellulose microfibrils form entanglements in their dispersions, which makes them similar in a certain rheological sense to the entangled solutions of linear macromolecules. Like polymer solutions, microfibrillar dispersions are capable of flowing with constant viscosity at low shear rates, while high shear rates reduce their effective viscosity to a certain level, until the flow of the dispersions ceases at very high shear rates due to tightening of the entanglement knots. However, microfibrillar dispersions have their own features due to their colloidal nature. They have a yield stress, the value of which is related to the stress required for deformation of the microfibrillar entanglement network, and therefore exhibit solid-like behavior of gels at small shear stresses below the yield stress.

In this way, cellulose microfibrils can provide liquid media with a yield stress and effectively increase their viscosity, i.e., they can act as a thickening agent. Moreover, microfibrils can do so in a non-volatile oil medium like triethyl citrate, suitable as a low-temperature base oil for greases. Furthermore, microfibrils make the oil effective viscosity less

temperature-dependent. All this gives the cellulose microfibrillar dispersions the potential to be used as fully biodegradable greases for low-temperature applications.

Remarkably, the results of tribological tests show that microfibrillated cellulose actually improves the antifriction and antiwear properties of triethyl citrate. This can occur by two mechanisms. In small quantities (1 %), microfibrils make the wear surface smooth by filling its defects. This reduces the time required for lapping of steel surfaces, significantly reduces their wear, and slightly decreases friction. In large quantities (7 %), microfibrils actively interact with the steel surface during lapping. In this case, the wear is quite high (at least for short observation times), but a tribological film is formed on the steel surface as a result. This film likely consists of a nanocellulose-based composite formed with the involvement of steel constituents undergoing joint chemical transformations with cellulose macromolecules under the influence of high local temperatures, pressures, and air oxygen. Due to its high smoothness, this tribological film significantly reduces friction of rubbing bodies, while their wear is quite high because of the strong abrasive action of microfibrils at their high concentration in the grease.

It seems that a moderately concentrated (5 %) dispersion of cellulose microfibrils is the best option for use as a grease. This dispersion combines both mechanisms of influence on the wear surface—filling its irregularities and creating a thin tribological film on it. Thanks to this, wear and friction are equally reduced, although not as well as they would be decreased individually if more diluted or more concentrated dispersions were applied.

5. Conclusion

The examination of dispersions of microfibrillated cellulose in triethyl citrate as low-temperature greases revealed that the tribological behavior of semi-dilute and concentrated systems is not the same despite the general similarity in their rheology. Nanocellulose flocs supposedly form a percolation structure in semi-dilute dispersions, giving them a yield stress gel-like behavior, significantly reducing wear of rubbing bodies by filling their surface irregularities, but only weakly decreasing the friction. In contrast, the gel structure in concentrated dispersions is formed by entanglements of cellulose microfibrils that notably wear down rubbing bodies but create a special tribological film on their surfaces, reducing the friction considerably as a result.

The limitations of the present study suggest challenges for further investigations. It is necessary to study the properties of nanocellulose greases at high temperatures that might be a problem in terms of thermo-oxidative degradation of cellulose and triethyl citrate and increased volatility of the latter. For a similar reason of oxidative and mechanical degradation of cellulose particles and macromolecules, it is essential also to carry out long-term tests of the greases in real working units. A separate issue is the biodegradability of the cellulose-based greases, both after and during operation, which is also worth investigating. Lastly, a study of friction and wear should be done at low temperatures directly, since the low-temperature fluidity alone might not be sufficient for high antiwear and antifriction grease properties.

Statement of originality

The submitted manuscript has not been published previously, is not under consideration for publication elsewhere, and, if accepted, will not be published elsewhere in the same form, in English or in any other language, including electronically.

CRediT authorship contribution statement

S.O. Ilyin: Conceptualization, Methodology, Investigation, Writing – original draft; **S.N. Gorbacheva:** Investigation, Writing – original draft; **A.Y. Yadykova:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

SEM images were obtained in the Shared Research Center "Analytical center of deep oil processing and petrochemistry of TIPS RAS". This research was carried out within the State Program of A.V. Topchiev Institute of Petrochemical Synthesis.

References

[1] Mannekote JK, Kailas SV, Venkatesh K, Kathyayini N. Environmentally friendly functional fluids from renewable and sustainable sources-a review. Renew Sustain Energy Rev 2018;81:1787–801. https://doi.org/10.1016/j.rser.2017.05.274.
[2] Qiu X, Hu S. "Smart" materials based on cellulose: a review of the preparations, properties, and applications. Materials 2013;6(3):738–81. https://doi.org/10.3390/ma6030738.
[3] Haigler CH. Biosynthesis and biodegradation of cellulose. CRC Press; 1990. https://doi.org/10.1201/9781482276985.
[4] Heinze T, Koschella A. Solvents applied in the field of cellulose chemistry: a mini review. Polímeros 2005;15:84–90. https://doi.org/10.1590/S0104-14282005000200005.
[5] de Oliveira Barud HG, da Silva RR, da Silva Barud H, Tercjak A, Gutierrez J, Lustri WR, et al. A multipurpose natural and renewable polymer in medical applications: bacterial cellulose. Carbohydr Polym 2016;153:406–20. https://doi.org/10.1016/j.carbpol.2016.07.059.
[6] Nazir MS, Wahjoedi BA, Yussof AW, Abdullah MA. Eco-friendly extraction and characterization of cellulose from oil palm empty fruit bunches. BioResources 2013;8:2161–72. https://doi.org/10.15376/biores.8.2.2161-2172.
[7] Srivastava N, Srivastava M, Mishra PK, Ramteke PW, Singh RL. New and future developments in microbial biotechnology and bioengineering: from cellulose to cellulase: strategies to improve biofuel production. Elsevier; 2019. https://doi.org/10.1016/C2018-0-01234-7.
[8] Gupta VK, Carrott PJM, Singh R, Chaudhary M, Kushwaha S. Cellulose: a review as natural, modified and activated carbon adsorbent. Bioresour Technol 2016;216:1066–76. https://doi.org/10.1016/j.biortech.2016.05.106.
[9] Li K, Zhang X, Du C, Yang J, Wu B, Guo Z, et al. Friction reduction and viscosity modification of cellulose nanocrystals as biolubricant additives in polyalphaolefin oil. Carbohydr Polym 2019;220:228–35. https://doi.org/10.1016/j.carbpol.2019.05.072.
[10] Gorbacheva SN, Yarmush YM, Ilyin SO. Rheology and tribology of ester-based greases with microcrystalline cellulose and organomodified montmorillonite. Tribol Int 2020;148:106318. https://doi.org/10.1016/j.triboint.2020.106318.
[11] Awang NW, Ramasamy D, Kadirgama K, Samykano M, Najafi G, Sidik NAC. An experimental study on characterization and properties of nano lubricant containing cellulose nanocrystal (CNC. Int J Heat Mass Transf 2019;130:1163–9. https://doi.org/10.1016/j.ijheatmasstransfer.2018.11.041.
[12] Li J, Lin N, Du C, Ge Y, Amann T, Feng H, et al. Tribological behavior of cellulose nanocrystal as an eco-friendly additive in lithium-based greases. Carbohydr Polym 2022;290:119478. https://doi.org/10.1016/j.carbpol.2022.119478.
[13] Tonkonogov BP, Kilyakova AY, Stenina ND, et al. Effect of thickener nature on properties of polyurealubricant compositions based on esters. Chem Technol Fuels Oils 2020;55(6):1–8. https://doi.org/10.1007/s10553-020-01083-0.
[14] Li J, Lin N, Du C, Ge Y, Amann T, Feng H, et al. Tribological behavior of cellulose nanocrystal as an eco-friendly additive in lithium-based greases. Carbohydr Polym 2022;290:119478. https://doi.org/10.1016/j.carbpol.2022.119478.
[15] Cunha AG, Gandini A. Turning polysaccharides into hydrophobic materials: a critical review. Part 1. Cellulose. Cellulose 2010;17(5):875–89. https://doi.org/10.1007/s10570-010-9434-6.
[16] Martín-Alfonso JE, López-Beltrán F, Valencia C, Franco JM. Effect of an alkali treatment on the development of cellulose pulp-based gel-like dispersions in vegetable oil for use as lubricants. Tribol Int 2018;123:329–36. https://doi.org/10.1016/j.triboint.2018.02.027.
[17] Cortés-Triviño E, Valencia C, Delgado MA, Franco JM. Rheology of epoxidized cellulose pulp gel-like dispersions in castor oil: influence of epoxidation degree and the epoxide chemical structure. Carbohydr Polym 2018;199:563–71. https://doi.org/10.1016/j.carbpol.2018.07.058.
[18] Gallego R, Cidade T, Sánchez R, Valencia C, Franco JM. Tribological behaviour of novel chemically modified biopolymer-thickened lubricating greases investigated in a steel–steel rotating ball-on-three plates tribology cell. Tribol Int 2016;94:652–60. https://doi.org/10.1016/j.triboint.2015.10.028.

[19] Núñez N, Martín-Alfonso JE, Valencia C, Sánchez MC, Franco JM. Rheology of new green lubricating grease formulations containing cellulose pulp and its methylated derivative as thickener agents. Ind Crops Prod 2012;37(1):500–7. https://doi.org/10.1016/j.indcrop.2011.07.027.

[20] Martín-Alfonso JE, Núñez N, Valencia C, Franco JM, Diaz MJ. Formulation of new biodegradable lubricating greases using ethylated cellulose pulp as thickener agent. J Ind Eng Chem (Amst, Neth) 2011;17(5–6):818–23. https://doi.org/10.1016/j.jiec.2011.09.003.

[21] Sánchez R, Franco JM, Delgado MA, Valencia C, Gallegos C. Thermal and mechanical characterization of cellulosic derivatives-based oleogels potentially applicable as bio-lubricating greases: influence of ethyl cellulose molecular weight. Carbohydr Polym 2011;83(1):151–8. https://doi.org/10.1016/j.carbpol.2010.07.033.

[22] Gorbacheva SN, Yadykova AY, Ilyin SO. Rheological and tribological properties of low-temperature greases based on cellulose acetate butyrate gel. Carbohydr Polym 2021;272:118509. https://doi.org/10.1016/j.carbpol.2021.118509.

[23] Borrero-López AM, Valencia C, Blánquez A, Hernández M, Eugenio ME, Franco JM. Cellulose pulp-and castor oil-based polyurethanes for lubricating applications: influence of streptomyces action on barley and wheat straws. Polymers 2020;12(12):2822. https://doi.org/10.3390/polym12122822.

[24] Gallego R, Arteaga JF, Valencia C, Franco JM. Rheology and thermal degradation of isocyanate-functionalized methyl cellulose-based oleogels. Carbohydr Polym 2013;98(1):152–60. https://doi.org/10.1016/j.carbpol.2013.04.104.

[25] Gallego R, Arteaga JF, Valencia C, Franco JM. Chemical modification of methyl cellulose with HMDI to modulate the thickening properties in castor oil. Cellulose 2013;20(1):495–507. https://doi.org/10.1007/s10570-012-9803-4.

[26] Gallego R, Arteaga JF, Valencia C, Franco JM. Thickening properties of several NCO-functionalized cellulose derivatives in castor oil. Chem Eng Sci 2015;134:260–8. https://doi.org/10.1016/j.ces.2015.05.007.

[27] Honary L, Richter E. Biobased lubricants and greases: technology and products. John Wiley & Sons; 2011. https://doi.org/10.1002/9780470971956.

[28] Brantseva TV, Ilyin SO, Gorbunova IY, Antonov SV, Korolev YM, Kerber ML. Epoxy reinforcement with silicate particles: rheological and adhesive properties-Part II: characterization of composites with halloysite. Int J Adhes Adhes 2016;68:248–55. https://doi.org/10.1016/j.ijadhadh.2016.04.005.

[29] Gorbacheva SN, Makarova VV, Ilyin SO. Hydrophobic nanosilica-stabilized graphite particles for improving thermal conductivity of paraffin wax-based phase-change materials. J Energy Storage 2021;36:102417. https://doi.org/10.1016/j.est.2021.102417.

[30] Kostyuk A, Ignatenko V, Smirnova N, et al. Rheology and adhesive properties of filled PIB-based pressure-sensitive adhesives. I. Rheology and shear resistance. J Adhes Sci Technol 2015;29(17):1831–48. https://doi.org/10.1080/01694243.2014.980616.

[31] Brantseva T, Antonov S, Kostyuk A, et al. Rheological and adhesive properties of PIB-based pressure-sensitive adhesives with montmorillonite-type nanofillers. Eur Polym J 2016;76:228–44. https://doi.org/10.1016/j.eurpolymj.2016.01.040.

[32] Yadykova AY, Ilyin SO. Rheological and adhesive properties of nanocomposite bitumen binders based on hydrophilic or hydrophobic silica and modified with bio-oil. Constr Build Mater 2022;342:127946. https://doi.org/10.1016/j.conbuildmat.2022.127946.

[33] Gorbacheva SN, Yadykova AY, Ilyin SO. A novel method for producing cellulose nanoparticles and their possible application as thickeners for biodegradable low-temperature greases. Cellulose 2021;28(16):10203–19. https://doi.org/10.1007/s10570-021-04166-1.

[34] Roman C, García-Morales M, Eugenio ME, Ibarra D, Martín-Sampedro R, Delgado MA. A sustainable methanol-based solvent exchange method to produce nanocellulose-based ecofriendly lubricants. J Clean Prod 2021;319:128673. https://doi.org/10.1016/j.jclepro.2021.128673.

[35] Clayton G, Clayton F. Patty's industrial hygiene and toxicology. 4th ed. New York NY: John Wiley & Sons; 1993–94. p. 3059.

[36] Einfeldt J, Kwasniewski A. Characterization of different types of cellulose by dielectric spectroscopy. Cellulose 2002;9(3):225–38. https://doi.org/10.1023/A:1021184620045.

[37] Moon RJ, Martini A, Nairn J, Simonsen J, Youngblood J. Cellulose nanomaterials review: structure, properties and nanocomposites. Chem Soc Rev 2011;40(7):3941–94. https://doi.org/10.1039/C0CS00108B.

[38] Kalia S, Dufresne A, Cherian BM, Kaith BS, Avérous L, Njuguna J, et al. Cellulose-based bio-and nanocomposites: a review. Int J Polym Sci 2011:837875. https://doi.org/10.1155/2011/837875.

[39] Siró I, Plackett D. Microfibrillated cellulose and new nanocomposite materials: a review. Cellulose 2010;17:459–94. https://doi.org/10.1007/s10570-010-9405-y.

[40] Mishra RK, Sabu A, Tiwari SK. Materials chemistry and the futurist eco-friendly applications of nanocellulose: status and prospect. J Saudi Chem Soc 2018;22(8):949–78. https://doi.org/10.1016/j.jscs.2018.02.005.

[41] Osong SH, Norgren S, Engstrand P. Processing of wood-based microfibrillated cellulose and nanofibrillated cellulose, and applications relating to papermaking: a review. Cellulose 2016;23(1):93–123. https://doi.org/10.1007/s10570-015-0798-5.

[42] Lindström T, Aulin C, Naderi A, Ankerfors M. Microfibrillated cellulose. Encycl Polym Sci Technol 2002:1–34. https://doi.org/10.1002/0471440264.pst614.

[43] Kumar V, Bollström R, Yang A, Chen Q, Chen G, Salminen P, et al. Comparison of nano-and microfibrillated cellulose films. Cellulose 2014;21(5):3443–56. https://doi.org/10.1007/s10570-014-0357-5.

[44] Lyadov AS, Maksimova YM, Ilyin SO, Gorbacheva SN, Parenago OP, Antonov SV. Specific features of greases based on poly-α-olefin oils with ureate thickeners of various structures. Russ J Appl Chem 2018;91:1735–41. https://doi.org/10.1134/S1070427218110010.

[45] Maksimova YM, Shakhmatova AS, Ilyin SO, Pakhmanova OA, Lyadov AS, Antonov SV, et al. Rheological and tribological properties of lubricating greases based on esters and polyurea thickeners. Pet Chem 2018;58(12):1064–9. https://doi.org/10.1134/S0965544118120071.

[46] Shafiei-Sabet S, Martinez M, Olson J. Shear rheology of micro-fibrillar cellulose aqueous suspensions. Cellulose 2016;23(5):2943–53. https://doi.org/10.1007/s10570-016-1040-9.

[47] Moberg T, Sahlin K, Yao K, Geng S, Westman G, Zhou Q, et al. Rheological properties of nanocellulose suspensions: effects of fibril/particle dimensions and surface characteristics. Cellulose 2017;24(6):2499–510. https://doi.org/10.1007/s10570-017-1283-0.

[48] Vadodaria SS, Onyianta AJ, Sun D. High-shear rate rheometry of micro-nanofibrillated cellulose (CMF/CNF) suspensions using rotational rheometer. Cellulose 2018;25(10):5535–52. https://doi.org/10.1007/s10570-018-1963-4.

[49] Ciftci GC, Larsson PA, Riazanova AV, Øvrebø HH, Wågberg L, Berglund LA. Tailoring of rheological properties and structural polydispersity effects in microfibrillated cellulose suspensions. Cellulose 2020;27(16):9227–41. https://doi.org/10.1007/s10570-020-03438-6.

[50] Koponen AI. The effect of consistency on the shear rheology of aqueous suspensions of cellulose micro-and nanofibrils: a review. Cellulose 2020;27(4):1879–97. https://doi.org/10.1007/s10570-019-02908-w.

[51] Gorbacheva SN, Ilyin SO. Morphology and rheology of heavy crude oil/water emulsions stabilized by microfibrillated cellulose. Energy Fuels 2021;35(8):6527–40. https://doi.org/10.1021/acs.energyfuels.0c02797.

[52] Yamagata Y, Niinobe S, Suga K, Nakano Y, Miyamoto K. Rheological and rheo-optical behaviors of nanocellulose suspensions containing unfibrillated fibers. Cellulose 2022;29(7):3703–19. https://doi.org/10.1007/s10570-022-04509-6.

[53] Pääkkö M, Ankerfors M, Kosonen H, Nykänen A, Ahola S, Österberg M, et al. Enzymatic hydrolysis combined with mechanical shearing and high-pressure homogenization for nanoscale cellulose fibrils and strong gels. Biomacromolecules 2007;8(6):1934–41. https://doi.org/10.1021/bm061215p.

[54] Attabi S, Mokhtari M, Taibi Y, Abdel-Rahman I, Hafez B, Elmsellem H. Electrochemical and tribological behavior of surface-treated titanium alloy Ti–6Al–4V. J Bio Tribo Corros 2019;5(1):1–11. https://doi.org/10.1007/s40735-018-0193-5.

[55] Fernández-Silva SD, Delgado MA, Roman C, García-Morales M. Rheological and tribological properties of nanocellulose-based ecolubricants. Nanomaterials 2021;11(11):2987. https://doi.org/10.3390/nano11112987.

[56] Ilyin S, Kulichikhin V, Malkin A. Characterization of material viscoelasticity at large deformations. Appl Rheol 2014;24:13653. https://doi.org/10.3933/applrheol-24-13653.

[57] Ilyin SO, Malkin AY, Kulichikhin VG. Application of large amplitude oscillatory shear for the analysis of polymer material properties in the nonlinear mechanical behavior. Polym Sci Ser A 2014;56:98–110. https://doi.org/10.1134/S0965545X14010039.

[58] Schramm G. A practical approach to rheology and rheometry. 2nd ed. Karlsruhe: Thermo Electron; 2004.

[59] Strelets LA, Ilyin SO. Effect of enhanced oil recovery on the composition and rheological properties of heavy crude oil. J Pet Sci Eng 2021;203:108641. https://doi.org/10.1016/j.petrol.2021.108641.

[60] Cousseau T. Grease lubrication: formulation effects on tribological performance. Tribology Mach Elem Fundam Appl 2021. https://doi.org/10.5772/intechopen.101549.

[61] Harnoy A. Bearing design in machinery: engineering tribology and lubrication. CRC Press; 2002. https://doi.org/10.1201/9780203909072.

[62] Lauri J, Haavisto S, Salmela J, Miettinen A, Fabritius T, Koponen AI. Online measurement of floc size, viscosity, and consistency of cellulose microfibril suspensions with optical coherence tomography. Cellulose 2021;28(6):3373–87. https://doi.org/10.1007/s10570-021-03745-6.

[63] Iotti M, Gregersen ØW, Moe S, Lenes M. Rheological studies of microfibrillar cellulose water dispersions. J Polym Environ 2011;19:137–45. https://doi.org/10.1007/s10924-010-0248-2.

[64] Malkin A, Kulichikhin V, Ilyin S. A modern look on yield stress fluids. Rheol Acta 2017;56(3):177–88. https://doi.org/10.1007/s00397-016-0963-2.

[65] Ilyin S, Roumyantseva T, Spiridonova V, Semakov A, Frenkin E, Malkin A, et al. Gels of cysteine/Ag-based dilute colloid systems and their rheological properties. Soft Matter 2011;7(19):9090–103. https://doi.org/10.1039/C1SM06007D.

[66] Ilyin SO, Malkin AY, Kulichikhin VG, Shaulov AY, Stegno EV, Berlin AA, et al. Rheological properties of polyethylene/metaboric acid thermoplastic blends. Rheol Acta 2014;53(5):467–75. https://doi.org/10.1007/s00397-014-0770-6.

[67] Malkin AY, Semakov AV, Kulichikhin VG. Modeling macromolecular movement in polymer melts and its relation to nonlinear rheology. Rheol Acta 2011;50(5–6):485. https://doi.org/10.1007/s00397-011-0556-z.

[68] Lauri J, Koponen A, Haavisto S, Czajkowski J, Fabritius T. Analysis of rheology and wall depletion of microfibrillated cellulose suspension using optical coherence tomography. Cellulose 2017;24(11):4715–28. https://doi.org/10.1007/s10570-017-1493-5.

[69] Malkin AY, Patlazhan SA. Wall slip for complex liquids–phenomenon and its causes. Adv Colloid Interface Sci 2018;257:42–57. https://doi.org/10.1016/j.cis.2018.05.008.

[70] Turpeinen T, Jäsberg A, Haavisto S, Liukkonen J, Salmela J, Koponen AI. Pipe rheology of microfibrillated cellulose suspensions. Cellulose 2020;27(1):141–56. https://doi.org/10.1007/s10570-019-02784-4.

[71] Malkin AY, Ilyin SO, Arinina MP, Kulichikhin VG. The rheological state of suspensions in varying the surface area of nano-silica particles and molecular weight of the poly (ethylene oxide) matrix. Colloid Polym Sci 2017;295(4):555–63. https://doi.org/10.1007/s00396-017-4046-4.
[72] Ilyin SO. Non-linearity in rheological properties of polymers and composites under large amplitude oscillatory shear. Polym Sci Ser A 2015;57(6):910–23. https://doi.org/10.1134/S0965545X15060103.
[73] Hyun K, Wilhelm M, Klein CO, Cho KS, Nam JG, Ahn KH, et al. A review of nonlinear oscillatory shear tests: analysis and application of large amplitude oscillatory shear (LAOS. Prog Polym Sci 2011;36(12):1697–753. https://doi.org/10.1016/j.progpolymsci.2011.02.002.
[74] Ilyin SO, Arinina MP, Malkin AY, Kulichikhin VG. Sol–gel transition and rheological properties of silica nanoparticle dispersions. Colloid J 2016;78(5):608–15. https://doi.org/10.1134/S1061933X16050070.
[75] Graessley WW. The entanglement concept in polymer rheology. In: The entanglement concept in polymer rheology. Berlin, Heidelberg: Springer; 1974. p. 1–179. https://doi.org/10.1007/BFb0031037.
[76] Cox WP, Merz EH. Correlation of dynamic and steady flow viscosities. J Polym Sci 1958;28(118):619–22. https://doi.org/10.1002/pol.1958.1202811812.
[77] Ilyin SO, Makarova VV, Polyakova MY, Kulichikhin VG. Phase behavior and rheology of miscible and immiscible blends of linear and hyperbranched siloxane macromolecules. Mater Today Commun 2020;22:100833. https://doi.org/10.1016/j.mtcomm.2019.100833.
[78] Gorbacheva SN, Ilyin SO. Structure, rheology and possible application of water-in-oil emulsions stabilized by asphaltenes. Colloids Surf A Physicochem Eng Asp 2021;618:126442. https://doi.org/10.1016/j.colsurfa.2021.126442.
[79] Ilyin SO, Makarova VV, Polyakova MY, Kulichikhin VG. Phase state and rheology of polyisobutylene blends with silicone resin. Rheol Acta 2020;59(6):375–86. https://doi.org/10.1007/s00397-020-01208-6.
[80] de Gennes PG. Scaling concepts in polymer physics. Ithaca: Cornell Univ. Press; 1979.
[81] Ilyin SO, Strelets LA. Basic fundamentals of petroleum rheology and their application for the investigation of crude oils of different natures. Energy Fuels 2018;32(1):268–78. https://doi.org/10.1021/acs.energyfuels.7b03058.
[82] Nifant'ev I, Bagrov V, Vinogradov A, et al. Methylenealkane-based low-viscosity ester oils: synthesis and outlook. Lubricants 2020;8(5):50. https://doi.org/10.3390/lubricants8050050.
[83] Kirkwood BR, Sterne JAC. Essential medical statistics. 2nd ed. Oxford: Blackwell Publishing; 2003.
[84] Esfahanian M, Hamrock BJ. Fluid-film lubrication regimes revisited. Tribol Trans 1991;34(4):628–32. https://doi.org/10.1080/10402009108982081.
[85] Nishiyama Y. Structure and properties of the cellulose microfibril. J Wood Sci 2009;55(4):241–9. https://doi.org/10.1007/s10086-009-1029-1.
[86] Cozza RC. Third abrasive wear mode: is it possible. J Mater Res Technol 2014;3(2):191–3. https://doi.org/10.1016/j.jmrt.2014.03.010.
[87] Tomków J, Czupryński A, Fydrych D. The abrasive wear resistance of coatings manufactured on high-strength low-alloy (HSLA) offshore steel in wet welding conditions. Coatings 2020;10(3):219. https://doi.org/10.3390/coatings10030219.
[88] Dong H. Tribological properties of titanium-based alloys. In: Surface engineering of light alloys. Woodhead Publishing; 2010. p. 58–80. https://doi.org/10.1533/9781845699451.1.58.
[89] Kolev M, Drenchev L, Petkov V. Wear analysis of an advanced Al–Al2O3 composite infiltrated with a tin-based alloy. Metals 2021;11(11):1692. https://doi.org/10.3390/met11111692.
[90] Awang NW, Ramasamy D, Kadirgama K, Najafi G, Sidik NAC. Study on friction and wear of Cellulose Nanocrystal (CNC) nanoparticle as lubricating additive in engine oil. Int J Heat Mass Transf 2019;131:1196–204. https://doi.org/10.1016/j.ijheatmasstransfer.2018.11.128.
[91] Malkin AY, Semakov AV, Kulichikhin VG. Self-organization in the flow of complex fluids (colloid and polymer systems): part 1: experimental evidence. Adv Colloid Interface Sci 2010;157(1–2):75–90. https://doi.org/10.1016/j.cis.2010.04.002.
[92] Efremenko VG, Shimizu K, Noguchi T, Efremenko AV, Chabak YG. Impact–abrasive–corrosion wear of Fe-based alloys: influence of microstructure and chemical composition upon wear resistance. Wear 2013;305(1–2):155–65. https://doi.org/10.1016/j.wear.2013.06.006.
[93] Delgado MA, Cortés-Triviño E, Valencia C, Franco JM. Tribological study of epoxide-functionalized alkali lignin-based gel-like biogreases. Tribol Int 2020;146:106231. https://doi.org/10.1016/j.triboint.2020.106231.
[94] Foley RT, Peterson MB, Zapf C. Frictional characteristics of cobalt, nickel, and iron as influenced by their surface oxide films. ASLE Trans 1963;6(1):29–39. https://doi.org/10.1080/05698196308971996.
[95] Clark WT, Pritchard C, Midgley JW. Mild wear of unlubricated hard steels in air and carbon dioxide. Proc Inst Mech Eng, Conf Proc 1967;182(14):97–106. https://doi.org/10.1243/PIME_CONF_1967_182_411_02.
[96] Vanysek, P. , Electrochemical series. In: Rumble, J R, editor. CRC handbook of chemistry and physics, 98th ed. Boca Raton, FL: CRC Press/Taylor & Francis; 2018. p 5–77; 5–83.